

KUBERNETES IN PRODUCTION · ISSUE 08 OF 08

Helm, cluster upgrades and the production checklist.

Boring upgrades are designed, not lucky.

08

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Swipe →

Helm values that scale

01 Sane defaults

values.yaml works for most environments.

02 Small overrides

values-prod.yaml holds only differences.

03 values.schema.json

Fail fast on typos and wrong types.

04 No secrets

Reference them; never inline them.

Deploy and roll back

BASH

```
helm upgrade --install api ./chart \
  -f values.yaml -f values-prod.yaml \
  --atomic --wait --timeout 10m
```

```
helm history api
helm rollback api 12
```

--atomic rolls back automatically if the release fails.



Node pool upgrade

- 1 Check deprecated APIs
- 2 Upgrade control plane
- 3 Surge or blue/green node pool
- 4 PDBs respected
- 5 Verify workloads
- 6 Remove old pool

Control planes move one minor version at a time. Plan the path.

Upgrade habits

DO

- ✓ Upgrade lower environments first
- ✓ Scan for removed APIs
- ✓ Use maintenance windows
- ✓ Keep a rollback node pool

AVOID

- ✗ Skip minor versions
- ✗ PDB minAvailable = replicas
- ✗ Upgrade during a release
- ✗ Ignore deprecation warnings

Production checklist

- ☒ Requests and limits set
- ☒ Probes configured
- ☒ PDB defined
- ☒ 2+ replicas across zones
- ☒ NetworkPolicy applied
- ☒ Runs as non-root
- ☒ Image pinned by digest
- ☒ Metrics, logs and alerts
- ☒ Rollback tested

KEY TAKEAWAY

Structure the chart, rehearse the upgrade, check the list.

- Small env overrides, schema-validated values
- --atomic releases with history
- Upgrade one minor version at a time

YOUR TURN

What is on your production checklist that is not on mine?

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Series complete — save it for your next review

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